

8(a)	<i>any three from:</i> 1 variation (with)in, ecosystems / habitats ; 2 number / variety, of (different) species ; 3 relative abundance of each species ; 4 genetic variation within each species ;	3
------	---	----------

Question	Answer	Marks																												
8(b)(i)	<table><tr><th>species</th><th>number</th><th>$\frac{n}{N}$</th><th>$\left(\frac{n}{N}\right)^2$</th></tr><tr><td><i>Rana temporaria</i></td><td>10</td><td>0.042</td><td>0.002</td></tr><tr><td><i>Leucorrhinia dubia</i></td><td>35</td><td>0.148</td><td>0.022</td></tr><tr><td><i>Hydrometra stagnorum</i></td><td>50</td><td>0.212</td><td>0.045</td></tr><tr><td><i>Lymnaea stagnalis</i></td><td>44</td><td>0.186</td><td>0.035</td></tr><tr><td><i>Gammarus pulex</i></td><td>97</td><td>0.411</td><td>0.169</td></tr><tr><td>Total</td><td>236</td><td></td><td>0.273</td></tr></table> n/N column correct ; (n/N)² column correct ; allow ecf Simpson’s Index of Diversity = 0.727 ; allow ecf	species	number	$\frac{n}{N}$	$\left(\frac{n}{N}\right)^2$	<i>Rana temporaria</i>	10	0.042	0.002	<i>Leucorrhinia dubia</i>	35	0.148	0.022	<i>Hydrometra stagnorum</i>	50	0.212	0.045	<i>Lymnaea stagnalis</i>	44	0.186	0.035	<i>Gammarus pulex</i>	97	0.411	0.169	Total	236		0.273	3
species	number	$\frac{n}{N}$	$\left(\frac{n}{N}\right)^2$																											
<i>Rana temporaria</i>	10	0.042	0.002																											
<i>Leucorrhinia dubia</i>	35	0.148	0.022																											
<i>Hydrometra stagnorum</i>	50	0.212	0.045																											
<i>Lymnaea stagnalis</i>	44	0.186	0.035																											
<i>Gammarus pulex</i>	97	0.411	0.169																											
Total	236		0.273																											
8(b)(ii)	(relatively) high value / close(r) to 1 (than 0) ; indicates (fairly) high (species) diversity ; allow ecf from 8(b)(i)	2																												